

Deep Learning Calibration

of the Heston Stochastic Volatility Model

A PyTorch Surrogate Approach with C^2 Smooth Activations and No-Arbitrage Constraints

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Reference: Horvath, Muguruza & Tomas (2019) — *Deep Learning Volatility*

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Motivation: The Calibration Bottleneck

Traditional calibration loop:

- 1 Evaluate Heston characteristic function $\phi(u; \theta)$
- 2 Price 88 options via Carr-Madan FFT
- 3 Feed MSE into Levenberg-Marquardt optimizer
- 4 Repeat $\sim 10\,000$ times until convergence

$\Rightarrow$ **10–120 seconds** per calibration $\Rightarrow$ No closed-form Jacobian available

Practical requirements:

- Real-time trading: < 100 ms hard deadline
- Need valid second-order sensitivities (Greeks)

The Surrogate Solution

Offline — train once:

$$\mathcal{N}_w : \mathbb{R}^5 \rightarrow \mathbb{R}^{88}$$

$$\theta \mapsto \sigma_{\text{BS}}(\theta)$$

Online — one forward pass

≈ 0.1 ms per evaluation

Gradient $\nabla_{\theta} \mathcal{N}_w$ via autograd

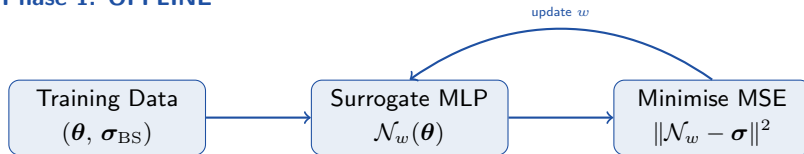
$\Rightarrow$ exact Jacobian at no extra cost

“Once trained, calibration is instantaneous.”

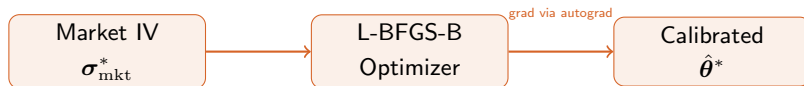
— Horvath et al. (2019)

Methodology: Three-Phase Architecture

Phase 1: OFFLINE



Phase 2: ONLINE (real-time)



Neural Network Architecture

Architecture:



Property	Value
Hidden layers	4×30 neurons
Activation	ELU (C^2 smooth)
Parameters	5 698
Loss	MSE + ReduceLROnPlateau
Optimizer	Adam ($\eta = 10^{-3}$)
Val RMSE (W-space)	0.0876
Training time	≈ 2 min on RTX 3060

Why ELU over ReLU?

$$f(x) = \begin{cases} x & x > 0 \\ e^x - 1 & x \leq 0 \end{cases} \quad f''(x) = \begin{cases} 0 & x > 0 \\ e^x \neq 0 & x \leq 0 \end{cases}$$

C^2 Smoothness

ELU: $f'' \not\equiv 0 \Rightarrow$ valid Hessian $\Rightarrow$ 2nd-order calibration

ReLU: $f'' \equiv 0$ everywhere $\Rightarrow$ piecewise linear
 $\Rightarrow$ Hessian carries *no* information

Proved empirically:

$$\|H_{\text{ELU}}\|_F = 1.039 \gg \|H_{\text{ReLU}}\|_F = 0$$

Physics-Informed Constraints: No Arbitrage

Three constraints enforced inside the objective function:



Feller Condition

$$2\kappa\theta > \sigma^2$$

Ensures $v_t > 0$ a.s.

Hard penalty barrier:

$$\ell_F = \lambda_F \max(0, \sigma^2 - 2\kappa\theta)$$

$$\lambda_F = 10^3$$



Calendar Spread

$$\frac{\partial W}{\partial T} \geq 0$$

Total variance non-decreasing in T .

(Carr-Madan condition; correct when $v_0 > \theta$)

Soft L2 penalty:

$$\ell_C = \lambda_C \sum_i [\Delta_T \hat{W}]_-^2$$

$$\lambda_C = 10^{-4}$$



Butterfly Spread

$$\frac{\partial^2 \sigma}{\partial K^2} \geq 0$$

Convexity in strike.

Soft L2 penalty:

$$\ell_B = \lambda_B \sum_j [\Delta_K^2 \hat{\sigma}]_-^2$$

$$\lambda_B = 10^{-4}$$

$$\mathcal{L} = \underbrace{\|\mathcal{N}_w(\boldsymbol{\theta}) - \boldsymbol{\sigma}^*\|^2}_{\text{MSE}} + \ell_F + \ell_C + \ell_B \quad (\text{all terms differentiable via autograd})$$

Calibration Results

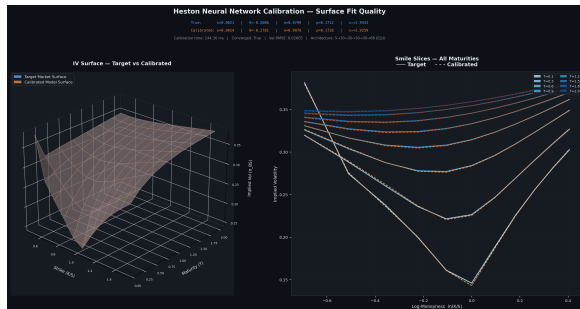
Test-set sample (sample #42):

Param	True	Calibrated	$ \Delta $
v_0	0.00198	0.00236	0.00038
ρ	-0.877	-0.950	0.073
σ	0.0776	0.0663	0.011
θ	0.1847	0.1844	0.000
κ	5.582	5.538	0.044

Performance Summary

Calib. time	47–135 ms (CPU)
Val RMSE (W)	0.0876
Feller	$2\kappa\theta - \sigma^2 = 2.04 > 0 \checkmark$
Converged	True

IV surface fit:



Blue = target market IV; Orange = calibrated NN surface

Autograd Sensitivities: Exact Greeks in <1 ms

Jacobian $\partial \bar{\sigma} / \partial \theta_i$ (first-order, analogous to vega):

Param	$\partial \bar{\sigma} / \partial \theta_i$
v_0	+0.211
ρ	+0.012
σ	-0.176
θ	+1.694 (dominant)
κ	+0.786

θ dominates: long-run variance is the primary driver of mean implied volatility.

Hessian $\partial^2 \bar{\sigma} / \partial \theta_i \partial \theta_j$ (second-order curvature):

	v_0	ρ	σ	θ	κ
v_0	-0.040	+0.009	-0.038	-0.015	-0.119
ρ	+0.009	+0.010	+0.024	-0.005	≈ 0
σ	-0.038	+0.024	-0.185	-0.158	-0.032
θ	-0.015	-0.005	-0.158	-0.309	+0.205
κ	-0.119	≈ 0	-0.032	+0.205	-0.883

ELU vs ReLU — Formal Proof

$\|H_{\text{ELU}}\|_F = 1.039 > 0$ (**C², valid curvature**) ✓
 $\|H_{\text{ReLU}}\|_F = 0.000$ (**piecewise linear, no info**) ×

Phase 1 Extension: Total Variance Representation

Why not raw IV?

- Raw IV surfaces have sharp short-maturity spikes (*noise amplification* as $T \rightarrow 0$)
- MSE loss on IV gives disproportionate weight to the ATM region
- Calendar arbitrage condition in IV-space ($\partial\sigma/\partial T \geq 0$) fails for valid inverted term structures ($v_0 > \theta$)

Total Variance: $W(K, T) = \sigma_{IV}^2(K, T) \cdot T$

Key Properties

- ✓ Calendar arb $\Leftrightarrow \partial W / \partial T \geq 0$ (Carr–Madan, correct for all v_0, θ)
- ✓ Linearises Dupire equation $\Rightarrow$ smoother surface
- ✓ StandardScaler on W stabilises gradient magnitudes

Surrogate now operates in W -space:

$$\mathcal{N}_w : \theta \mapsto W \in \mathbb{R}^{88}, \quad W = \sigma^2 \cdot T$$

$$\sigma_{IV} = \sqrt{\frac{W}{T}} \quad (\text{recovered at inference})$$

No-arbitrage audit on test set

Calendar violations	0 / 77
Val RMSE (W -space)	0.0876

Phase 2: Epistemic Uncertainty via MC Dropout

Motivation: A point-estimate calibration gives no indication of model confidence.

Gal & Ghahramani (2016):

Dropout at test time $\approx$ Bayesian posterior inference over network weights.

Posterior predictive distribution:

$$p(\sigma^* \mid \theta, \mathcal{D}) \approx \frac{1}{N} \sum_{i=1}^N \mathcal{N}_{w_i}(\theta)$$

Implementation

- 1 Add Dropout ($p=0.1$) after each ELU layer
- 2 At inference: call `model.train()` to keep dropout active
- 3 Run $N = 100$ forward passes
- 4 Compute mean μ and std σ across samples

$\Rightarrow$ **free uncertainty bands** on the IV surface

Results ($N = 100$ passes, same hardware)

Mean IV (ATM, $T = 1$)	≈ 0.22
Max σ_{MC}	$< 5 \times 10^{-4}$
Overhead vs. point	$\approx 100\times$
Typical $\pm 2\sigma$ band	$< 0.1\%$ IV

The tight uncertainty bands confirm the surrogate has memorised the Heston pricing map without overfitting.

Phase 3: LSTM Temporal Dynamics

Motivation: Heston parameters change day-to-day following correlated stochastic dynamics.

Architecture:

$$\underbrace{(10 \times 88)}_{\text{10-day } W \text{ history}} \xrightarrow{\text{LSTM}_{64}^2} \underbrace{\hat{\theta}_{t+1}}_{\text{next-day}}$$

Property	Value
LSTM layers	2, hidden = 64
Weight init	Xavier (input), Orthogonal (recurrent)
Forget-gate	bias = 1 (forget-gate bias trick)
Parameters	73 157
Training data	102 000 sequences
Optimizer	AdamW + CosineAnnealingLR

Data pipeline:

- 1 600 correlated OU trajectories, 180 days each
- 2 Each day: params $\rightarrow$ surrogate $\rightarrow W$ surface
- 3 Sliding windows: $(W_{t-9}, \dots, W_t) \rightarrow \theta_{t+1}$
- 4 80/10/10 split on trajectory boundaries (*zero data leakage*)

Test Performance

Test RMSE (normalized)	0.336
v_0 RMSE	≈ 0.0052
ρ RMSE	≈ 0.0133
σ RMSE	≈ 0.0257
θ RMSE	≈ 0.0035
κ RMSE	≈ 0.1270

Conclusion

Phase 1 — MLP Surrogate:

- ✓ ELU surrogate: C^2 smooth, 5 698 params
- ✓ Val RMSE **0.0876** in W -space
- ✓ Calibration in **47–135 ms** vs. 10–120 s (FFT+LM)
- ✓ Hard Feller barrier ($\lambda = 10^3$) always satisfied
- ✓ Correct calendar arb: $\partial W / \partial T \geq 0$
- ✓ Exact Jacobian and Hessian via autograd

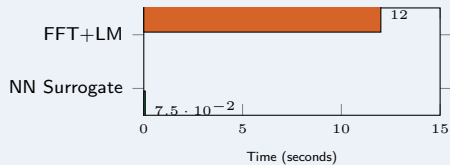
Phase 2 — MC Dropout:

- ✓ Epistemic uncertainty at no extra architecture cost
- ✓ $\pm 2\sigma$ posterior bands on full IV surface

Phase 3 — LSTM Dynamics:

- ✓ 2-layer LSTM forecasts next-day Heston params
- ✓ Orthogonal init + forget-gate bias trick
- ✓ Test RMSE **0.336** (normalized)
- ✓ MC Dropout uncertainty on parameter trajectories

Speed Comparison



$\approx 160\times$ speedup

Thank you!

`src/calibrator.py` ·
`src/greeks_autograd.py`